Sebastian Forero Duque

Data Engineering | Process Automation (IBM Cloud Pak) | Backend
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PROFESSIONAL PROFILE

Systems and Computer Engineering student (pending degree, UPB) with hands-on experience developing data pipelines, ETL processes, databases, REST API integration, and cloud infrastructure (AWS). During my professional internship at XM S.A. E.S.P. (Grupo ISA), I automated data flows and enterprise architecture processes across the Cybersecurity, Applications, Infrastructure, Interoperability, and Data layers, using IBM Cloud Pak for Business Automation (CP4BA) and IBM Business Automation Studio. Experience building predictive and descriptive Machine Learning models with Python, Scikit-learn, TensorFlow, Knime, and Weka. English C1 certified by EF Education First.

PROFESSIONAL EXPERIENCE

IT Intern - Process Automation

XM S.A. E.S.P. (Grupo ISA) | *Medellin, Colombia* | *January-July 2026*

Built the end-to-end automation of the Technical Architecture Boards process using IBM Business Automation Studio (BAW) within IBM Cloud Pak for Business Automation (CP4BA), covering the full session lifecycle from request to commitment tracking.

- Designed and built dynamic forms and business logic in BAW to manage the complete architecture board workflow.
- Developed data handling and integration logic for structured storage of requests, decisions, and commitments.
- Connected the platform with internal APIs to trigger automatic notifications, keeping stakeholders informed in real time.
- Supported and optimized the room-booking process for architecture sessions, reducing manual coordination effort.
- Built Power BI dashboards from session metrics, providing visibility into meeting volume, scheduling efficiency, and commitment tracking for management decision-making.

RELEVANT PROJECTS

Air Quality Dashboard - ETL and Data Visualization

UPB / SIATA | *Medellin, Colombia* | *2025*

- Built a complete ETL pipeline to extract, transform, and load environmental sensor data from SIATA; converted PM2.5 measurements into an Air Quality Index (AQI) according to EPA standards.
- Developed an interactive geospatial dashboard in Streamlit with Pandas, Matplotlib, and map layers, with real-time filters by zone and time range.
- Demonstrated end-to-end Data Engineering skills: ingestion, transformation, storage, and visualization.

Emotion Detection Model - Predictive Modeling with ML

UPB | *Medellin, Colombia* | *2024*

- Developed a deep learning predictive model with Convolutional Neural Networks (CNN) using TensorFlow and Keras for real-time emotion classification from video.
- Optimized hyperparameters and applied data augmentation to reduce the false positive rate and improve model generalization.
- Performed descriptive analysis of the training data distribution to identify class imbalance and improve model fairness.

Supplier Evaluation Platform - Backend

UPB / CTIC | *Medellin, Colombia* | *2024*

- Led a development team building a full-stack application with Java (Spring Boot), MySQL, and Angular to automate supplier evaluation under ICONTEC regulations.
- Designed the relational database schema and stored procedures for data integrity and auditability.

REST API - Chemical Compound Management

UPB | *Medellin, Colombia* | *2024*

- Built a RESTful API with Spring Boot and PostgreSQL including stored procedures and complex relational queries

for managing chemical compounds and elements.

TECHNICAL SKILLS

Data Engineering & ETL: Pandas, ETL pipeline design, data ingestion, data transformation, data validation, SQL, PostgreSQL, MySQL, MongoDB, NoSQL, data modeling, data warehousing concepts

Programming Languages: Python, Java, JavaScript, C#

Cloud & Infrastructure: AWS (EC2, S3), Docker, IBM Cloud Pak for Business Automation (CP4BA), IBM Business Automation Studio

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, predictive modeling, descriptive modeling, CNN, neural networks, Transformers, NLP (basic), model training and evaluation, hyperparameter tuning

Visualization & Analysis: Streamlit, Matplotlib, Seaborn, Pandas, Jupyter Notebook, Tableau, Power BI

Backend & APIs: Spring Boot, REST API development and consumption

Automation & Architecture: BPMN modeling, IBM Business Automation Studio, ArchiMate, UML

Other Tools: Git, GitHub, Docker, Postman, n8n

EDUCATION

Systems and Computer Engineering (Degree Pending)
Universidad Pontificia Bolivariana (UPB) | *Medellin, Colombia* | *2022 - Present*

English C1 Certification
EF Education First | *Brighton, England* | *2024*

English B2 Certificate
Centro Colombo Americano | *Medellin, Colombia* | *2012 - 2021*

High School Diploma with Honors
Colegio de la UPB | *Medellin, Colombia* | *2010 - 2021*

LANGUAGES

Spanish: Native English: C1 - EF Education First (2024)